

Alvaro Javier Vargas Guerrero

PHD STUDENT · VUB

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Federated Learning, Causality, Causal Discovery, Federated Causal Discovery, Kernel Techniques

Education

Vrije Universiteit Brussel

PHD ARTIFICIAL INTELLIGENCE

- Promotor: Prof. Dr. Guy Nagels
- Promotor: Prof. Dr. Lynn Houthuys

Brussels, Belgium

July 2025 - present

Vrije Universiteit Brussel

MASTER APPLIED SCIENCES AND ENGINEERING (84%)

- Promotor Master's Thesis: Prof. Dr. Ann Nowé

Brussels, Belgium

September 2022 - July 2024

Professional Experience

2024-2025 **IT Consultant**, Addestino

2022-2023 **Software Developer**, Vrije Universiteit Brussel

Publications

PUBLISHED

Vargas Guerrero, A. J., Dillen, A., Loeckx, J., & Nagels, G. *A Systematic Review of Federated Causal Discovery: Definitions and Methods.*, FLTA 2026 Main Track

Vargas Guerrero, A. J., Wang, X., & Doan, Q. M., & Nagels, G. *Beyond Weights and Gradients: A Taxonomy of Federated Learning Messages.*, FL@FM Workshop at IJCAI 2026

IN REVIEW

Vargas Guerrero, A. J., Houthuys, L., & Nagels, G. *COSMIC: Scalable Causal Discovery via Nyström Macro-Tier Peeling*

PRE-PRINTS

Wang, X., Nguyen, T. L., Chén, O. Y., Vargas Guerrero, A. J., Doan, Q. M., Hu, T., Denissen, S., Van Schependom, J., & Nagels, G. *Federated Adversarial Network (FAN): A Generative Proxy-Based, Defense-Enhanced, and Non-IID-Robust Federated Learning Method.*

Teaching Experience

2026 **Statistical Foundations of Machine Learning**, Teaching Assistant

2026 **Arts in de Maatschappij**, Teaching Assistant

2026 **Current Trends in Artificial Intelligence**, Teaching Assistant

Mentoring

2026 **Kamila Maria Obregon Ortega**, Internship on Federated Causal Discovery

2026 **Yaun Francis**, Internship on Federated Causal Discovery

2027 **Mohammad Amman Ul Haq Gadda**, Master Thesis on fMRI Causality

2027 **Houda Yousfi**, Master Thesis on Federated Causal Discovery Benchmarks

2027 **Srivasini Hari**, Master Thesis on Federated Deep Causal Discovery